

Miao Li

Postdoctoral Researcher in NLP/AI
School of Informatics, The University of Edinburgh

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WORKING EXPERIENCE

Postdoctoral Researcher, The University of Edinburgh
Institute for Language, Cognition and Computation
Supervised by Prof. [Mirella Lapata](#)

Mar. 2025 – Present

EDUCATION BACKGROUND

Visiting Ph.D. The University of Edinburgh
Institute for Language, Cognition and Computation
Advised by Prof. [Mirella Lapata](#)

Mar. 2024 – Mar. 2025

Ph.D. The University of Melbourne
School of Computing and Information Systems
Advised by Prof. [Eduard Hovy](#) and Prof. [Jey Han Lau](#)

Dec. 2020 – Mar. 2025

M.Sc. (Exam-exempt Postgraduate) Chinese Academy of Sciences
Institute of Software, Chinese Academy of Sciences

Aug. 2017 – July 2020

B.S. Northeastern University
School of Computer Science and Technology

Aug. 2013 – July 2017

RESEARCH INTERESTS

- Complex reasoning: Enabling models to integrate and reason over voluminous, heterogeneous, dispersed information in complex contexts
- Reasoning-centric post-training: Designing large-scale, experience-driven learning paradigms that draw inspiration from human cognition to acquire complex reasoning capabilities
- Model evaluation: Developing rigorous evaluation methodologies to measure, characterize, and understand reasoning capabilities and behaviors

TECHNICAL SKILLS

- Proficient in problem formulation, large-scale data curation, modelling and evaluation methodologies for advanced natural language reasoning and generation tasks
- Experienced in post-training and experimentation with large language models (LLMs), especially for tasks involving complex contexts (e.g., multi-document question answering, and long-context understanding)
- Skilled in Python and PyTorch, with hands-on experience in large-scale model training and efficient inference of LLMs (leveraging modern frameworks such as Transformers, OpenRLHF and vLLM)
- Solid theoretical foundation in natural language processing, deep learning and fundamental statistical machine learning, with particular emphasis on language modelling, text generation and distributed computing

RESEARCH PUBLICATIONS

- [SciTrek: Evaluating and Improving Long-Context Numerical Reasoning over Scientific Articles](#) arXiv'26
Miao Li, Alexander Gurung, Irina Saparina, and Mirella Lapata.
Preprint (July 2026).
- [Long-Context Reasoning Through Proxy-Based Chain-of-Thought Tuning](#) ACL'26
Miao Li, Irina Saparina, Alexander Gurung, and Mirella Lapata.
Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics
- [Decomposed Opinion Summarization with Verified Aspect-Aware Modules](#) ACL'25
Miao Li, Jey Han Lau, Eduard Hovy, Mirella Lapata
Findings of the Association for Computational Linguistics: ACL 2025.
- [NewsBench: A Systematic Evaluation Framework for Assessing Editorial Capabilities of Large Language Models in Chinese Journalism](#) ACL'24

Miao Li, Ming-Bin Chen, Bo Tang, Shengbin Hou, Pengyu Wang, Haiying Deng, Zhiyu Li, Feiyu Xiong, Keming Mao, Peng Cheng, and Yi Luo

Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics

- [A Sentiment Consolidation Framework for Meta-Review Generation](#) ACL'24

Miao Li, Jey Han Lau, Eduard Hovy

Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics

- [Summarizing Multiple Documents with Conversational Structure for Meta-review Generation](#) EMNLP'23

Miao Li, Eduard Hovy, Jey Han Lau

Findings of the Association for Computational Linguistics: EMNLP 2023

- [DeltaScore: Fine-grained Story Evaluation with Perturbations](#) EMNLP'23

Zhuohan Xie, **Miao Li**, Trevor Cohn, Jey Han Lau

Findings of the Association for Computational Linguistics: EMNLP 2023

- [Compressed Heterogeneous Graph for Abstractive Multi-document Summarization](#) AAAI'23

Miao Li, Jianzhong Qi, Jey Han Lau

Proceedings of the 37th AAI Conference on Artificial Intelligence

- [Personalized Abstractive Opinion Tagging](#) SIGIR'22

Mengxue Zhao, Yang Yang, **Miao Li**, Jingang Wang, Wei Wu, Pengjie Ren, Maarten de Rijke and Zhaochun Ren

Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval

- [A Benchmark and Comprehensive Survey on Knowledge Graph Entity Alignment via Representation Learning](#) VLDBJ'22

Rui Zhang, Bayu Distiawan Trisedy, **Miao Li**, Yong Jiang, Jianzhong Qi

The International Journal on Very Large Data Bases

- [A Topic Augmented Text Generation Model: Joint Learning of Semantics and Structural Features](#) EMNLP'19

Hongyin Tang, **Miao Li**, Beihong Jin

Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing

- [A New Effective Neural Variational Model with Mixture-of-Gaussians Prior for Text Clustering](#) ICTAI'19

Miao Li, Hongyin Tang, Beihong Jin, Chengqing Zong

2019 IEEE 31st International Conference on Tools with Artificial Intelligence

- [Clustering Large-Scale Origin-Destination Pairs: A Case Study for Public Transit in Beijing](#) UIC'18

Miao Li, Beihong Jin, Hongyin Tang, Fusang Zhang

The 15th IEEE International Conference on Ubiquitous Intelligence and Computing

RESEARCH GRANTS

- Google Gemini Academic Program Award, USD50,000.00 (GCP credits) 2026 - 2027

HONORS & AWARDS

- Google Conference Scholarship, Google LLC Jan. 2023
- AAI-23 Student Scholarship, Association for the Advancement of Artificial Intelligence Jan. 2023
- National Scholarship for Graduates (top 2%), Education Ministry of China Nov. 2018
- The First Prize Scholarship (top 8%), University of Chinese Academy of Sciences Oct. 2017 & 2018
- Excellent Student Cadre, University of Chinese Academy of Sciences Jun. 2018 & 2019
- Merit Student, University of Chinese Academy of Sciences Jun. 2018 & 2019
- Outstanding Graduate, Northeastern University Jun. 2017
- Outstanding Graduate Thesis Award (1/272), Northeastern University Jul. 2017
- National Scholarship for Undergraduates (top 2%), Education Ministry of China Oct. 2016
- The First Prize Scholarship (top 4%), Northeastern University Sep. 2015 & 2016
- Excellent Student Cadre, Northeastern University Oct. 2015 & 2016

TEACHING EXPERIENCES

- [Foundations of Natural Language Processing](#) (INFR10078) Course Tutor
AY 2025/2026 (Semester 2), The University of Edinburgh

PROFESSIONAL SERVICES

- Journal reviewer: TKDE 2025

- Reviewer committee:
NeurIPS 2026, ICLR 2026, ACL 2023-2026, EMNLP 2023-2026, NAACL 2025, EACL 2023, NLP CC 2022-2023, WSDM 2022, SIGIR 2022, SIGKDD 2022, CIKM 2022
- Reviewer for ACL Rolling Review (2021 – Present)